

Model Tuning, Regularization & Reproducible Pipelines

1. Search Summary

Three models were evaluated with a common preprocessing pipeline (feature engineering → ColumnTransformer → classifier), 5-fold Stratified CV, random_state=42 fixed throughout, and Precision as the optimization target. Baseline cross-validated scores (default hyperparameters) were compared against RandomizedSearchCV results (n_iter=50 per model, scoring="precision").

Model	Baseline CV Precision	Tuned CV Precision	Δ
Logistic Regression	0.7464	0.7520	+0.0056
Random Forest	0.7123	0.8334	+0.1211
HistGradientBoosting	0.7831	0.8701	+0.0870

HistGradientBoosting was selected: highest tuned precision (0.8701), with best parameters learning_rate=0.01, max_depth=8, max_iter=100, l2_regularization=0.1. Tuning gave the largest lift to Random Forest and HistGradientBoosting; Logistic Regression barely moved, consistent with the diagnostics.

2. Bias/Variance Diagnostics

- **Learning curve (best model):** Train/validation gap shrinks from 0.067 (2.9k rows) to ~0.01–0.017 (≥14k rows); validation precision is flat (0.865–0.879) from ~6.6k rows onward. Reading: low-variance, near bias-limited — more data alone will not meaningfully raise precision further; better features would help more.
- **Logistic Regression — effect of C:** Train/validation gap stays under 0.0035 across C = 0.001 to 100; validation precision peaks at ~0.749 near C≈0.167. Reading: Logistic Regression never overfits in this range — its ~0.75 ceiling is a model-capacity (bias) limit, not a regularization problem. Further C tuning is not the lever to pull.
- **HistGradientBoosting — effect of max_depth:** Gap grows from 0.002 (depth 3) to 0.009 (depth 8) to 0.012 (unrestricted); validation precision peaks at depth 8 (0.8701) and dips slightly beyond it even as train precision keeps climbing. Reading: depth 8 is the correct stopping point — confirms the value RandomizedSearchCV selected independently.

Concrete fixes, ranked by expected payoff: (1) invest in feature engineering over collecting more rows — the learning curve has plateaued; (2) keep max_depth=8 — deeper adds variance with no validation gain; (3) do not rely on C-tuning to close Logistic Regression's gap to the tree ensemble — it is a capacity limit.

3. Calibration & Threshold Selection

- **Calibration:** Brier Score improved from 0.1126 (raw) to 0.0993 (isotonic-calibrated) — calibration curve moved closer to the diagonal.
- **Threshold:** Business requirement: minimum precision of 0.89 (false positives costlier than missed positives). The lowest threshold clearing that floor was chosen, rather than the threshold with maximum precision — unconstrained max-precision degenerates toward predicting almost no positives.
- **Selected threshold:** 0.70 → Precision 0.894, Recall 0.401, F1 0.554 on the test set. This is a deliberate tradeoff: the model misses ~60% of true positives to stay precise. A lower floor (e.g. 0.85 → threshold 0.65, recall 0.455) would recover more recall if preferred.

4. Final Hold-Out Test Performance

Metric	Value
Accuracy	0.8398
Precision	0.8937
Recall	0.4014

F1	0.5540
ROC-AUC	0.9106
PR-AUC	0.7983
Brier Score	0.0993

Evaluated once, on the untouched hold-out test set, using the final calibrated pipeline at threshold 0.70. Strong ranking ability (ROC-AUC 0.911, PR-AUC 0.798) and well-calibrated probabilities (Brier 0.099) support the precision-first, recall-conservative production profile intended by the threshold choice.

5. Production Notes

- **Artifact:** preprocessing + calibration + threshold are bundled into a single joblib artifact; inference requires only raw feature rows.
- **Expected behavior:** expect the model to flag positives it is highly confident about (precision ≈ 0.89) while under-flagging genuine high-income cases (recall ≈ 0.40). This is by design for the stated business requirement, not a defect.